

Mitochondrial Research: Yeast and Human Cells as Models.

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The evolution of complex eukaryotes would have been impossible without mitochondria, key cell organelles responsible for the oxidative metabolism of sugars and the bulk of ATP production [...].

Transmission of mitochondrial DNA following assisted reproduction and nuclear transfer.

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Mitochondria are the organelles responsible for producing the majority of a cell's ATP and also play an essential role in gamete maturation and embryo development. ATP production within the mitochondria is dependent on proteins encoded by both the nuclear and the mitochondrial genomes, therefore co-ordination between the two genomes is vital for cell survival. To assist with this co-ordination, cells normally contain only one type of mitochondrial DNA (mtDNA) termed homoplasmy. Occasionally, however, two or more types of mtDNA are present termed heteroplasmy. This can result from a combination of mutant and wild-type mtDNA molecules or from a combination of wild-type mtDNA variants. As heteroplasmy can result in mitochondrial disease, various mechanisms exist in the natural fertilization process to ensure the maternal-only transmission of mtDNA and the maintenance of homoplasmy in future generations. However, there is now an increasing use of invasive oocyte reconstruction protocols, which tend to bypass mechanisms for the maintenance of homoplasmy, potentially resulting in the transmission of either form of mtDNA heteroplasmy. Indeed, heteroplasmy caused by combinations of wild-type variants has been reported following cytoplasmic transfer (CT) in the human and following nuclear transfer (NT) in various animal species. Other techniques, such as germinal vesicle transfer and pronuclei transfer, have been proposed as methods of preventing transmission of mitochondrial diseases to future generations. However, resulting embryos and offspring may contain mtDNA heteroplasmy, which itself could result in mitochondrial disease. It is therefore essential that uniparental transmission of mtDNA is ensured before these techniques are used therapeutically.

Hydrogenosomes in the diplomonad *Spironucleus salmonicida*.

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Acquisition of the mitochondrion is a key event in the evolution of the eukaryotic cell, but diversification of the organelle has occurred during eukaryotic evolution. One example of such mitochondria-related organelles (MROs) are hydrogenosomes, which produce ATP by substrate-level phosphorylation with hydrogen as a byproduct. The diplomonad parasite *Giardia intestinalis* harbours mitosomes, another type of MRO. Here we identify MROs in the salmon parasite *Spironucleus salmonicida* with similar protein import and Fe-S cluster assembly machineries as in *Giardia* mitosomes. We find that hydrogen production is prevalent in the diplomonad genus *Spironucleus*, and that *S. salmonicida* MROs contain enzymes characteristic of hydrogenosomes. Evolutionary analyses of known hydrogenosomal components indicate their presence in the diplomonad ancestor, and subsequent loss in *Giardia*. Our results suggest that hydrogenosomes are metabolic adaptations predating the split between parabasalids and diplomonads, which is deeper than the split between animals and fungi in the eukaryotic tree.

The hydrogenosome peripheral vesicle: similarities with the endoplasmic reticulum.

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The hydrogenosome, an organelle that produces molecular hydrogen and ATP from the oxidation of pyruvate or malate under anaerobic conditions, presents some characteristics common to mitochondria. The hydrogenosome of *Tritrichomonas foetus*, a cattle parasite, is a spherical organelle that presents a peripheral vesicle the origin and behavior of which is poorly known. In this article it is reported an ultrastructural and microanalytical study using energy dispersive X-ray analysis, 3D reconstruction and cytochemistry of the hydrogenosome peripheral vesicle and then compare the results with the endoplasmic reticulum and the nuclear envelope of *T. foetus*. Similarities between the hydrogenosome peripheral vesicle and the ER are presented. This study included: (1) the detection of ER enzymes by cytochemistry, such as glucose-6-phosphatase, IDPase, acid phosphatase and Ca^{2+} -ATPase; (2) elemental composition by X-ray microanalysis and the mapping of calcium, phosphorus and oxygen in both ER and hydrogenosome peripheral vesicle; (3) freeze-fracture; (4) TEM of routine and cryofixed cells by high-pressure freezing and freeze-substitution; (5) 3D reconstruction, (6) monoclonal antibody anti-trichomonads ER; and (6) other cytochemical techniques that detects ER, such as the ZIO and lectins. We found a similar composition of the tested enzymes and other elements present in the ER when compared with the hydrogenosome's peripheral vesicle. It was concluded that, like mitochondria, hydrogenosome presents relationships with the ER, especially the peripheral vesicle.

Regulation of Mitochondrial Ca^{2+} Uptake.

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Mitochondria are responsible for ATP production but are also known as regulators of cell death, and mitochondrial matrix Ca^{2+} is a key modulator of both ATP production and cell death. Although mitochondrial Ca^{2+} uptake and efflux have been studied for over 50 years, it is only in the past decade that the proteins responsible for mitochondrial Ca^{2+} uptake and efflux have been identified. The identification of the mitochondrial Ca^{2+} uniporter (MCU) led to an explosion of studies identifying regulators of the MCU. The levels of these regulators vary in a tissue- and disease-specific manner, providing new insight into how mitochondrial Ca^{2+} is regulated. This review focuses on the proteins responsible for mitochondrial transport and what we have learned from mouse studies with genetic alterations in these proteins.

Glycolysis, but not Mitochondria, responsible for intracellular ATP distribution in cortical area of podocytes.

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Differentiated podocytes, a type of renal glomerular cells, require substantial levels of energy to maintain glomerular physiology. Mitochondria and glycolysis are two major producers of ATP, but the precise roles of each in podocytes remain unknown. This study evaluated the roles of mitochondria and glycolysis in differentiated and differentiating podocytes. Mitochondria in differentiated podocytes are located in the

central part of cell body while blocking mitochondria had minor effects on cell shape and migratory ability. In contrast, blocking glycolysis significantly reduced the formation of lamellipodia, a cortical area of these cells, decreased the cell migratory ability and induced the apoptosis. Consistently, the local ATP production in lamellipodia was predominantly regulated by glycolysis. In turn, synaptopodin expression was ameliorated by blocking either mitochondrial respiration or glycolysis. Similar to differentiated podocytes, the differentiating podocytes utilized the glycolysis for regulating apoptosis and lamellipodia formation while synaptopodin expression was likely involved in both mitochondrial OXPHOS and glycolysis. Finally, adult mouse podocytes have most of mitochondria predominantly in the center of the cytosol whereas phosphofructokinase, a rate limiting enzyme for glycolysis, was expressed in foot processes. These data suggest that mitochondria and glycolysis play parallel but distinct roles in differentiated and differentiating podocytes.

Reactive Nitrogen Species in Mitochondria and Their Implications in Plant Energy Status and Hypoxic Stress Tolerance.

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Hypoxic and anoxic conditions result in the energy crisis that leads to cell damage. Since mitochondria are the primary organelles for energy production, the support of these organelles in a functional state is an important task during oxygen deprivation. Plant mitochondria adapted the strategy to survive under hypoxia by keeping electron transport operative even without oxygen via the use of nitrite as a terminal electrons acceptor. The process of nitrite reduction to nitric oxide (NO) in the mitochondrial electron transport chain recycles NADH and leads to a limited rate of ATP production. The produced ATP alongside with the ATP generated by fermentation supports the processes of transcription and translation required for hypoxic survival and recovery of plants. Non-symbiotic hemoglobins (called phyoglobins in plants) scavenge NO and thus contribute to regeneration of NAD(+) and nitrate required for the operation of anaerobic energy metabolism. This overall operation represents an important strategy of biochemical adaptation that results in the improvement of energy status and thereby in protection of plants in the conditions of hypoxic stress.

Organelle Stress and Crosstalk in Kidney Disease.

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Organelles play important roles in maintaining cellular homeostasis. Organelle stress responses, especially in mitochondria, endoplasmic reticula (ER), and primary cilia, are deeply involved in kidney disease pathophysiology. Mitochondria are the center of energy production in most eukaryotic cells. Renal proximal tubular cells are highly energy demanding and abundant in mitochondria. Mitochondrial dysfunctions in association with energy metabolism alterations produce reactive oxygen species and promote inflammation in proximal tubular cells, resulting in progression of kidney disease. The ER play critical roles in controlling protein quality. Unfolded protein response (UPR) pathways are the adaptive response to ER stress for maintaining protein homeostasis. UPR pathway dysregulation under pathogenic ER stress often occurs in glomerular and tubulointerstitial cells and promotes progression of kidney disease. The primary cilia sense extracellular signals and maintain calcium homeostasis in cells. Dysfunction of the primary cilia in autosomal dominant polycystic kidney disease reduces the calcium

concentration in proximal tubular cells, leading to increased cell proliferation and retention of cyst fluid. In recent years, the direct interaction at membrane contact sites has received increased attention in association with the development of imaging technologies. The part of the ER that is directly connected to mitochondria is termed the mitochondria-associated ER membrane (MAM), which regulates calcium homeostasis and phospholipid metabolism in cells. Disruption of MAM integrity collapses cellular homeostasis and leads to diseases such as diabetes and Alzheimer disease. This review summarizes recent research on organelle stress and crosstalk, and their involvement in kidney disease pathophysiology. In addition, potential treatment options that target organelle stress responses are discussed.

The role of mitochondrial-related nuclear genes in age-related common disease.

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Mitochondria are critical organelles that provide energy as ATP to the cell. Besides 37 genes encoded by mitochondrial genome, it has been estimated that over 1500 nuclear genes are required for mitochondrial structure and function. Thus, mutations of many genes in the nuclear genome cause dysfunction of mitochondria that can lead to many severe conditions. Mitochondrial dysfunction often results in reduced ATP synthesis, higher levels of reactive oxygen species (ROS), imbalanced mitochondrial dynamics, and other detrimental effects. In addition to rare primary mitochondrial disorders, these mitochondrial-related genes are often associated with many common diseases. For example, in neurodegenerative diseases such as Parkinson's, Alzheimer's, and Huntington disease, mitochondrial energy metabolism abnormalities can greatly affect brain function. Cancer cells are also known to exhibit repressed mitochondrial ATP production in favor of glycolysis, which fuels the aggressive proliferation and metastasis of tumor tissues, leading many to speculate on a possible relationship between compromised mitochondrial function and cancer. The association between mitochondrial dysfunction and diabetes is also unsurprising, given the organelle's crucial role in cellular energy utilization. Here, we will discuss the multiple lines of evidence connecting mitochondrial dysfunction associated with mitochondria-related nuclear genes to many of the well-known disease genes that also underlie common disease.

Mitochondrial interaction with the endosomal compartment in endocytosis and mitochondrial transfer.

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Mitochondria are essential organelles required for cellular processes ranging from energy production to cellular differentiation. To perform these functions, mitochondria physically and functionally interact with other organelles such as the endoplasmic reticulum (ER) and endosomes. While the role of ER-mitochondria contact sites is well established, the interaction between mitochondria and endosomes has only recently been reported. These interactions are involved in lipid and ion transfer and potentially play a crucial role in mitochondria quality control and the release of mitochondrial components within extracellular vesicles. Here, we will discuss the current view of mitochondria-endosome interaction, both physically and functionally.